Towards LLM-Centric Multimodal Fusion: A Survey on Integration Strategies and Techniques

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Abstract

The rapid progress of Multimodal Large Language Models(MLLMs) has transformed the AI landscape. These models combine pre-trained LLMs with various modality encoders. This integration requires a systematic understanding of how different modalities connect to the language backbone. Our survey presents an LLM-centric analysis of current approaches. We examine methods for transforming and aligning diverse modal inputs into the language embedding space. This addresses a significant gap in existing literature. We propose a classification framework for MLLMs based on three key dimensions. First, we examine architectural strategies for modality integration. This includes both the specific integration mechanisms and the fusion level. Second, we categorize representation learning techniques as either joint or coordinate representations. Third, we analyze training paradigms, including training strategies and objective functions. By examining 125 MLLMs developed between 2021 and 2025, we identify emerging patterns in the field. Our taxonomy provides researchers with a structured overview of current integration techniques. These insights aim to guide the development of more robust multimodal integration strategies for future models built on pre-trained foundations.

1 Introduction

MLLMs represent a significant breakthrough in AI research. These models combine foundation LLMs with specialized encoders for different modalities including images, audio, and video (Caffagni et al., 2024; Wu et al., 2023). Models that integrate multiple modalities achieve richer context comprehension by leveraging complementary information from different sources (Shukang Yin et al., 2023).

Prior studies: Existing surveys provide valuable insights into multimodal LLMs. Shukang Yin et al. (2023) conducted a comprehensive review covering various aspects of multimodal LLMs including components, datasets, and training methodologies, but focused less on modality integration approaches. Similarly, Song et al. (2025) attempted to categorize modality integration methods into multimodal convertor and multimodal perceiver, but this rigid classification creates confusion when architectural components serve multiple functional roles depending on implementation context. For example, an MLP(Multi-Layer Perceptron) layer might be used by Liu et al. (2024) to project features into the LLM embedding space, while Wang et al. (2024a) use it to reduce the number of image tokens. This gap is particularly evident in architectural mechanisms, representation techniques, and training methodologies. A detailed comparison with prior surveys appears in Appendix A(Table 2).

Key challenge: Although the proliferation of MLLMs, a significant challenge persists in understanding the varied functional roles of common architectural components. Identical modules are often employed with differing contextual intentions across models. This forces researchers to invest considerable time in identifying the intended purpose of components, hindering the efficient design and adaptation of MLLMs.

Contributions: Our survey addresses this gap by analyzing 125 MLLM papers(2021-2025). We introduce a structured framework examining MLLMs through three key dimensions: 1) architectural strategies for modality fusion(including mechanisms such as Abstraction, Projection, Semantic Embedding, and Cross-attention layers, along with fusion levels like early, intermediate, or hybrid), 2) representation learning paradigms(joint or coordinate representation), and 3) training methodologies(training strategies and objective functions).

Our primary contributions include:

- We propose a novel cross-modality fusion

Model	Abstractor Layer	Projection Layer	Semantic Embedding Layer	Cross-attention Layer
AnomalyGPT ★(Gu et al., 2024)	-	Linear	Convolution	-
BLIP-2 ★(Li et al., 2023b)	-	Linear	Q-former	-
Cambrian-1 ♦(Tong et al., 2024)	-	-	Cross-attention (like Convolution)	-
CogAgent ♢ (Hong et al., 2024)	-	MLP	-	Within Model
Flamingo ♦ (Alayrac et al., 2022)	Perceiver Resampler	-	-	Within Model
InstructBLIP (Dai et al., 2023)	-	Linear	Q-former	-
Kosmos-1 (Huang et al., 2023)	Perceiver Resampler	MLP (last layer of ViT)	-	-
LHRS-Bot (Muhtar et al., 2024)	-	-	Perceiver Resampler	-
LLaMA-Adapter V2 ♦ (Gao et al., 2023)	-	Linear	-	Within Model (self attention layer)
LLaVA (Liu et al., 2023)	-	Linear	-	-
LLaVA-1.5 (Liu et al., 2024)	-	MLP	-	-
MM1 (McKinzie et al., 2024)	C-Abstractor	-	-	-
mPLUG-Owl2 ♦ (Ye et al., 2024c)	Visual Abstractor	-	-	Modality Adaptive Module
mPLUG-Owl3 ♦ (Ye et al., 2024a)	-	Linear	-	Within Model (Hyper Attention Transformer block)
PMC-VQA (Zhang et al., 2024d)	-	MLP / Transformer	-	-
SEAL ▲(Sun et al., 2025)	Convolution	MLP	-	-
SEED ★(Ge et al., 2023)	-	Linear	Q-former	-
VideoChat (K.-Y. R. Li et al., 2023)	Q-former	Linear	-	-
VILA (Lin et al., 2024b)	-	Linear / Transformer	-	-
VisionLLM (Wang et al., 2023)	-	Transformer	-	-
X-InstructBLIP (Panagopoulou et al., 2024)	-	Q-former + Linear	-	-

Table 1: Selected Comparisons of MLLMs. Key: Fusion Level is indicated by ♦ (Intermediate) or ♢ (Hybrid); other fusion instances are 'Early'. Representation is indicated by ▲ (Coordinate) or ★ (Hybrid); other representation instances are 'Joint'.

mechanisms taxonomy that explains how the same architectural components can perform different contextual functions based on researchers' design intentions.

- We provide a cross-modality fusion mechanism table 1 that presents existing MLLM studies at a glance according to contextual fusion mechanisms and fusion levels.

- We identify emerging patterns and design principles from existing MLLM research to guide future multimodal system development.

This taxonomy provides researchers with a structured overview of current integration techniques for building MLLMs on pre-trained foundations.

The remainder of this paper is organized as follows. Section 2 provides essential background on foundational concepts. Section 3 presents our classification taxonomy for architectural strategies. Section 4 examines representation learning approaches. Section 5 investigates training methodologies. Sections 6, 7, and 8 discuss future challenges, conclusions, and limitations respectively.

2 Background

Multimodality Multimodality refers to combining different data types(text, images, audio, etc.) within a single AI system (Ngiam et al., 2011; Wu et al., 2023; Shukang Yin et al., 2023; Li and Tang, 2024). Humans understand the world through multiple senses, and similarly, AI systems perform better when using different input types together. When a model processes multiple modalities at once, it can capture information that single-modality models would miss (Ngiam et al., 2011; Karpathy and Fei-Fei, 2015). For example, MLLMs combine visual and textual data to understand images in language terms (Alayrac et al., 2022; Peng Wang et al., 2022; Liu et al., 2023; Li et al., 2023b; OpenAI et al., 2024a). This integration enables better performance on tasks like image captioning and visual question answering by connecting language with visual perception (Agrawal et al., 2017; Driess et al., 2023; OpenAI et al., 2024a).

Large Language Model LLMs form the foundation of MLLMs by providing strong language capabilities learned from massive text datasets (Radford, 2018; Radford et al., 2019; Brown et al., 2020). These transformer-based models typically contain billions of parameters and excel at reasoning and few-shot learning (Brown et al., 2020; Chowdhery et al., 2023; Wei et al., 2022; Touvron et al., 2023; Gunasekar et al., 2023; Jiang et al., 2023; Team et al., 2025). In most MLLMs, the pre-trained LLM remains frozen or lightly tuned to preserve its knowledge while reducing training costs (Tsimpoukelli et al., 2021; Li et al., 2023b; Liu et al., 2023). This approach leverages existing LLMs(like LLaMA or Vicuna) as reasoning engines without

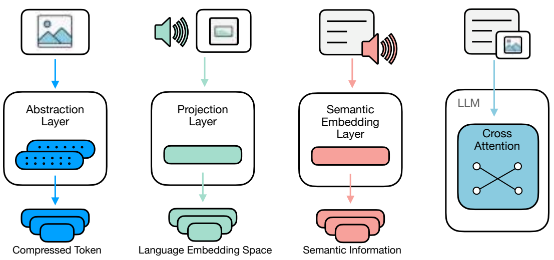

Figure 1: Proposed contextual fusion mechanisms

full finetuning, making multimodal system development more efficient (Touvron et al., 2023; Peng et al., 2023a; Shukang Yin et al., 2023).

Feature Encoder MLLMs use specialized encoders to convert modality inputs into vector representations. For images, Vision Transformers(ViT) or convolutional networks extract visual features (Krizhevsky et al., 2012; He et al., 2016; Dosovitskiy et al., 2021), while models like Wav2Vec 2.0 encode audio signals (Schneider et al., 2019; Baevski et al., 2020). Many successful MLLMs use encoders with multimodal pretraining, such as CLIP, which aligns images and text in a shared embedding space (Radford et al., 2021). These pre-trained encoders provide features already aligned with language concepts, making it easier for the LLM to understand them. The modular design of MLLMs incorporates specialized encoders for different modalities while keeping them frozen to preserve their expertise (Liu et al., 2023; Peng Wang et al., 2022; Zhang et al., 2023a; K.-Y. R. Li et al., 2023). Recent work on universal encoders, like ImageBind, aims to create common embedding spaces for multiple modalities, potentially simplifying future MLLM architectures (Girdhar et al., 2023).

3 Architectural strategies for Modality fusion

This section examines the key mechanisms for integrating non-text modalities with large language models. Understanding these integration mechanisms is crucial because researchers should able to select appropriate integration strategies based on their targeted purpose.

3.1 Cross-modality fusion mechanisms

We propose a taxonomy of four contextual fusion mechanisms based on our analysis of recent multimodal LLM literature: **Projection**, **Abstraction**, **Semantic embedding** and **Cross-attention**. Figure 1 illustrates the concept of the mechanisms. These mechanisms bridge the gap between different data types, allowing LLMs to process and generate responses based on diverse inputs. These mechanisms can be implemented through various architectural components. However, each component may serve multiple purpose depending on researcher's intention. To address this confusion, this section examines how researchers implement these mechanisms to connect non-text features with language representations in the LLM's embedding space.

3.1.1 Abstraction layer

Abstraction layer is used to control the number of tokens from non-text modality features. When integrating non-text modalities with pre-trained LLMs, feature extraction using encoders is processed first. However, these extracted features undergo various processing to be used as input for LLMs. Abstraction layer acts as an information bottleneck(Mu et al., 2023) for extracted features, producing fewer or fixed-length tokens.

The abstraction process can mitigate several issues with extracted features. First, the number of features can vary depending on the data format. This variability in input feature count can increase the difficulty of model architecture design. Second, the number of features can be large. For example, a ViT encoder extracts features by dividing images into patches. Therefore, if the input image has high resolution, it results in numerous features. This large number of features can increase model size and computational costs during training and inference.

Huang et al. (2023); Laurençon et al. (2024); Li et al. (2023c); Xu et al. (2024c); Yue et al. (2024); Ye et al. (2024c) use Perceiver Resampler to achieve abstraction effects. Perceiver Resampler is a concept proposed in Flamingo(Alayrac et al., 2022), with a Perceiver-based structure(Jaegle et al., 2021). The main characteristic of Perceiver Resampler is using fixed-length learnable queries to perform cross-attention with input features, resulting in output tokens with a fixed length determined by the learnable queries.

Zhang et al. (2023b); Tang et al. (2024); Shao et al. (2024); Somepalli et al. (2024) utilize Q-former as an abstraction layer. Q-former is a structure proposed in the BLIP-2(Li et al., 2023b). Similar to Perceiver Resampler, Q-former uses learnable queries to fix the number of output tokens. The difference is that self-attention layers and cross-attention layers alternate. In self-attention layers, information exchange occurs between learnable queries, while in cross-attention layers, information exchange occurs between modality features and learnable queries.

MM1(McKinzie et al., 2024) uses C-Abstractor as an abstraction layer. C-Abstractor (Convolutional Abstractor) and D-Abstractor (Deformable attention-based Abstractor) are proposed by Honeybee(Cha et al., 2024). They pointed out that Perceiver Resampler and Q-former based abstraction can inherently suffer from a risk of information loss. To address the limitations, the architectures are proposed. (Yu et al., 2024; Sun et al., 2025) used simple convolution layers for abstraction. Convolution layers are expected to capture spatial or temporal dependencies within features.

3.1.2 Projection layer

Projection layers map extracted features into the language embedding space, making them more comprehensible to the LLM. Using projection layers reduces the need to update LLM parameters for modality integration, which is economical and preserves the LLM's pretrained knowledge. This advantage has led to their widespread adoption in research.

Gao et al. (2023); K.-Y. R. Li et al. (2023); Li et al. (2023a); Lin et al. (2024b); Liu et al. (2023); Su et al. (2023); Zhang et al. (2023b); Ge et al. (2023) use linear layers as projection layers. The simple structure of linear layers allows for cross-modality integration effects with minimal resources. Liu et al. (2024) mentions that linear layers may be insufficient for effective projection due to their simple structure. Lai et al. (2024); Liu et al. (2024); Wang et al. (2024b); Hong et al. (2024); Hu et al. (2024); Xu et al. (2024b); Shao et al. (2024); Jain et al. (2024); Chen et al. (2024a) use more sophisticated MLPs for projection compared to linear layers. Notably, LLaVA-1.5 (Liu et al., 2024) improved performance by changing the projection layer from a linear layer to an MLP. Lin et al. (2024b); Li et al. (2024a); Zhang et al. (2024d) use transformer architecture which can serve enough capacity for projection. Panagopoulou et al. (2024) use Q-former for instruction aware projection.

3.1.3 Semantic embedding layer

Beyond refining(by Abstraction Layer) or projecting extracted features, there are attempts to incorporate high-level information into features. Semantic embedding layers add high-level information to non-text features through various mechanisms.

Q-former is commonly used as a semantic embedding layer. The learnable queries used in Q-former can serve as instructions to extract semantic information from input features. Li et al. (2023b); Dai et al. (2023); Hu et al. (2024); Ren et al. (2024); Ge et al. (2023); He et al. (2024); Chen et al. (2024a); Qi et al. (2024); Mittal et al. (2024) use Q-former as a layer for high-level semantics. InstructBLIP (Dai et al., 2023) proposes a structure that explicitly includes text instructions as input to Q-former alongside learnable queries. Mu et al. (2023); Qi et al. (2024); Chen et al. (2024a); Hu et al. (2024, 2023) use Q-former proposed in InstructBLIP to more explicitly incorporate high-level semantics. Muhtar et al. (2024) use learnable queries in Perceiver Resampler to incorporate semantic information. Gu et al. (2024) uses convolution layers for semantic information. Tong et al. (2024) paper adds spatial inductive bias to vision features using learnable queries along with convolution-like cross-attention layers.

3.1.4 Cross-attention layer

A more explicit cross-modality integration method is utilizing cross-attention layers. Cross-attention layers allow LLMs to dynamically attend to non-text features. In cross-attention layers, query and key-value are generated from different modalities to enable inter-modality information exchange. Since most modern LLMs have transformer structures, they already contain cross-attention layers. Modality integration can be induced by feeding non-text features into these internal cross-attention layers.

Flamingo (Alayrac et al., 2022) induce modality integration by directly inputting non-text embeddings into internal cross-attention layers. Gong et al. (2023); Moor et al. (2023); Li et al. (2023c); Yue et al. (2024) follow similar structures based on Flamingo. CogVLM (Wang et al., 2024b) expands internal matrices for Query, Key, and Value for vision modality. CogAgent (Hong et al., 2024) improves high-resolution image processing performance of CogVLM by feeding hidden states from CLIP-based vision encoder into internal cross-attention layers. mPLUG-Owl2 (Ye et al., 2024c) receives both vision and text inputs in the model's internal attention layers and uses projection matrices to generate modality-specific keys and values. mPLUG-Owl3 (Ye et al., 2024a) proposes Hyper Attention Transformer blocks that perform text-only self-attention and vision-text cross-attention at the same level before integration through adaptive gates. Besides using existing internal cross-attention layers, modality integration can be attempted by adding external cross-attention layers. LLaMA-VID (Li et al., 2024c) adds separate cross-attention layers to compute cross-attention between text queries and vision embeddings.

3.2 Fusion level

We propose a taxonomy of fusion strategies in MLLMs based on fusion level, i.e. the stage at which non-text modalities are integrated relative to the core LLM. As illustrated in Figure 2, we distinguish Early Fusion, Intermediate Fusion, and Hybrid Fusion. These fusion levels are orthogonal to semantic fusion mechanisms(Projection, Abstraction, Cross-attention, Semantic Embedding): the same mechanism can be applied at different stages based on model design.

3.2.1 Early Fusion

Early Fusion merges non-text features before feeding them into the LLM, offering advantages in computational efficiency, model integrity preservation, and simplified integration. This approach processes and transforms modality embeddings into the LLM embedding space, reducing sequence length.

For instance, Kosmos-1 (Huang et al., 2023) merges patch tokens via an abstraction layer(Perceiver Resampler) to distill high-dimensional visual inputs into a compact representation. This early abstraction preserves salient information while reducing sequence length for computational efficiency. LLaVA (Liu et al., 2023) projects CLIP embeddings through a linear projection layer into token embeddings compatible with the LLM, maintaining the pretrained backbone and minimizing integration overhead. BLIP-2 (Li et al., 2023b) employs a Q-Former semantic embedding module to extract focused visual queries, enriching language generation with high-level context and filtering out background noise to improve multimodal alignment. LLaMA-VID (Li et al., 2024c) introduces an external cross-attention layer before LLM input to dynamically fuse projected vision tokens with text embeddings, enabling robust token alignment and dynamic feature selection while keeping the core LLM frozen.

3.2.2 Intermediate Fusion

Intermediate Fusion integrates non-text modalities within the LLM's transformer layers rather than before them. This approach allows for dynamic, layer-wise interaction between modalities throughout the processing pipeline.

For example, Flamingo (Alayrac et al., 2022) uses an abstraction layer(Perceiver Resampler) to distill image features and gated cross-attention layers to fuse them with text. By placing cross-attention mid-layer, Flamingo dynamically queries visual details at each generation step, improving contextual grounding and reducing reliance on static embeddings. CogVLM (Wang et al., 2024b) maps vision embeddings via an MLP projection layer before using in-model cross-attention adapters. This preserves the frozen LLM backbone while enabling precise visual queries at each layer, balancing parameter efficiency and fine-grained cross-modal interaction. Cambrian-1 (Tong et al., 2024) employs a convolutional semantic embedding layer via cross-attention, introducing spatial inductive bias directly into the LLM's layers. This enhances region-level alignment between visual features and text tokens, improving performance on spatial reasoning tasks. LLaMA-Adapter V2 (Gao et al., 2023) attaches lightweight, zero-initialized cross-attention adapters to a frozen LLaMA, achieving multimodality with under 1M new parameters. Embedding adapters inside transformer layers allows seamless modality mixing with minimal overhead and preserves pretrained knowledge. These intermediate fusion designs enable progressive, layer-wise modality merging, delivering precise grounding and task-specific reasoning with minimal model disturbance.

3.2.3 Hybrid Fusion

Hybrid Fusion combines the advantages of both Early and Intermediate fusion strategies, offering computational efficiency while maintaining fine-grained integration capabilities. This approach provides the preprocessing benefits of early fusion and the dynamic interaction benefits of intermediate fusion.

For example, CogAgent (Hong et al., 2024) first projects vision features through an MLP before feeding them into the LLM, then uses cross-

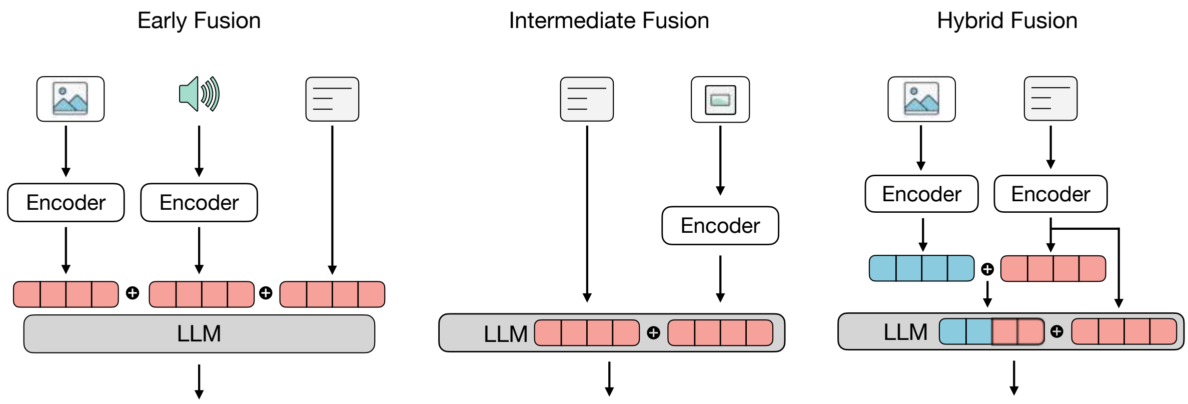

Figure 2: Proposed a taxonomy of LLM-centric fusion level(Early, Intermediate, and Hybrid)

attention adapters to inject detailed interface elements, combining global layout and fine detail. This design leverages efficient early alignment and precise mid-layer fusion, making it ideal for vision understanding. ManipLLM (Li et al., 2024a) injects cross-attention layers within transformer blocks to seamlessly merge modality tokens midstream. This semantic embedding approach preserves initial feature alignment while enabling dynamic, token-level integration—key for tasks that demand both global context and precise action grounding in real time. These hybrid systems combine the efficiency of early alignment with the precision of in-model fusion, offering both fast retrieval and deep, token-level integration for complex tasks.

4 Representation of Multi-Modal Data

The choice of representation learning on multimodal data—how heterogeneous inputs like text, images, audio, video, and sensor signals are integrated and processed—depends largely on the specific task at hand. In this section, we categorize multimodal representations into three classes: **Joint**, **Coordinated**, and **Hybrid**, each suited to different objectives and requirements.

We discuss how Multimodal LLMs adopt these paradigms, outlining representative methods, their respective advantages and limitations.

4.1 Joint Representation

Joint Representation merges features from all modalities into one shared space. Different inputs (text, images, audio, video) become tokens in the same Transformer. The model fuses them via self-attention and cross-attention.

There are two main approaches. **Projection layer**: A projection or adapter maps each modality's features into the LLM's embedding space. These projected tokens join the text token sequence. Models like LLaVA (Liu et al., 2023), MiniGPT-4 (Zhu et al., 2024a), LLaMA-Adapter (Zhang et al., 2024c), and others (Liu et al., 2024; Driess et al., 2023; Peng Wang et al., 2022; Wang et al., 2024d; Gao et al., 2023; Chen et al., 2023b; Lai et al., 2024; Zhang et al., 2024e; Lin et al., 2024b) use this method. This approach preserves the LLM's pre-trained structure and integrates modalities with minimal changes to the core model.

Cross-attention layer: The LLM is extended with cross-attention layers that let text queries attend to visual keys and values at selected transformer blocks. Flamingo (Alayrac et al., 2022), InstructBLIP (Dai et al., 2023), Qwen-VL (Bai et al., 2023), and similar models (Wang et al., 2023; Peng et al., 2023b; Zitkovich et al., 2023; Ye et al., 2024b; Zhu et al., 2024b; Zhao et al., 2024) use cross-attention for detailed grounding. This approach enables fine-grained, token-level fusion by directing the model's attention to relevant modality features at each generation step. Additionally, several systems, such as MoVA and MMICL, combine cross-attention with adapter layers to achieve their specific goals. Because all modalities interact within a single network, joint methods excel at fine-grained visual question answering, image captioning, and multimodal dialogue. Nevertheless, they can struggle when a modality is missing at inference time and may be inefficient for retrieval-oriented tasks where independent modality embeddings are advantageous.

4.2 Coordinated Representation

Coordinated Representation uses separate encoders for each modality and contrastive learning to align their outputs. Matching pairs(e.g., image–text) are pulled together, while mismatched pairs are pushed apart. CLIP-style training is the canonical example: an image encoder and a text encoder are optimized so that embeddings of matching pairs converge while mismatched pairs diverge. Recent variants such as HACL (Jiang et al., 2024) and DPE (Zhang et al., 2024a) extend this idea.

This approach builds a shared embedding space that supports fast cross-modal retrieval and zero-shot transfer. Encoders stay modular: they can be swapped or updated without modifying the LLM. It also handles missing modalities gracefully and scales to new data.

Choose coordinated representation when retrieval efficiency, encoder flexibility, and robustness are priorities. However, without a fusion module, it cannot capture fine-grained token-level grounding or complex reasoning. In those cases, a joint representation may be more effective.

4.3 Hybrid Representation

Hybrid Representation combines coordinated alignment and joint fusion in two steps. First, separate encoders extract modality embeddings and align them with a contrastive loss(e.g., CLIP-style). Second, a fusion module—such as a cross-attention Transformer block or Q-Former—merges these embeddings into a unified representation. Models like BLIP-2 (Li et al., 2022, 2023b), MoVA (Zong et al., 2024), and Video-LLaMA (Zhang et al., 2023b) follow this pattern.

This design keeps encoders modular for fast cross-modal retrieval and zero-shot transfer, while the fusion step enables fine-grained, token-level grounding for detailed reasoning. Choose hybrid representation when you need both efficient search and precise multimodal integration. If you only need deep fusion without retrieval, a joint representation may suffice; if you only need fast similarity search, a coordinated representation is simpler.

Selecting a representation depends on your data, inference setup, and task needs. Joint representation is best for deep feature fusion and fine-grained reasoning. Coordinated representation excels at fast retrieval and modular encoder updates. Hybrid representation offers both search efficiency and detailed grounding. Future MLLMs may dynamically switch or combine these modes based on input modality and task context. As models handle more modalities(audio, video, 3D, robotic data) under real-time demands, representation methods must stay efficient and scalable.

5 Training Strategies and Objectives

The training process for modality integration can be categorized as **Single-stage training**, **Two-stage training**, or **Multi-stage training** based on the number of training phases.

5.1 Single-stage training

Li et al. (2022); Tsimpoukelli et al. (2021); Dai et al. (2023); Su et al. (2023); Li et al. (2023c) uses a single training step to induce modality integration. Integrating modalities into LLMs requires enormous cross-modal datasets, and captioning datasets are commonly used for this purpose. However, simple pairwise datasets like captioning datasets may not be sufficient for modality integration (Zhu et al., 2024a). Therefore, mixing various datasets for training is also utilized (Zitkovich et al., 2023; Driess et al., 2023; Lai et al., 2024).

5.2 Two-stage training

To address the limitations of single-stage training, many approaches use multiple stages (Liu et al., 2024; Li et al., 2023b; Zhu et al., 2024a; Zhang et al., 2023b; Peng et al., 2023b; K.-Y. R. Li et al., 2023; Wang et al., 2024b; Gao et al., 2023). The 2-stage approach typically consists of an alignment stage and an instruction tuning stage. In the alignment stage, a large amount of pairwise captioning data is used to train alignment modules such as projection layers. To enhance instruction following capabilities that might be lacking in captioning dataset training, the second stage uses a smaller amount of sophisticated instruction datasets for instruction tuning.

The 2-stage approach offers several benefits. First, it allows the use of both large but simple datasets and small but sophisticated datasets. Second, it mitigates catastrophic forgetting. When integrating new modalities into LLMs, it's important to effectively utilize the LLM's pre-trained knowledge. However, catastrophic forgetting can occur during the process of updating LLM parameters. In most 2-stage approaches, alignment is performed while keeping the LLM frozen. This helps mitigate the catastrophic forgetting problem compared to techniques that update model parameters throughout the entire training process.

5.3 Multi-stage training

Some studies attempt multi-stage training by more precisely controlling the dataset and target components. Multi-stage training is used to change target components by stage (Bai et al., 2023), vary the datasets used by stage (Liu et al., 2025; Zhang et al., 2023a; Wang et al., 2024c; Chen et al., 2023a), or expand datasets stage by stage from a curriculum learning perspective (Li et al., 2023a; Gong et al., 2024; Muhtar et al., 2024; Chen et al., 2024b; Zhu et al., 2024b).

5.4 Training Objectives

To integrate non-text modalities, MLLMs are trained with a language modeling loss. This loss measures how well the model predicts the next text token given previous tokens and multimodal features. It is typically implemented as cross-entropy or negative log-likelihood over the token distribution. By using this objective, the MLLM learns both fluent language generation and alignment with non-text inputs.

Most MLLM research uses this LM objective, though many works also combine it with various losses for better modality alignment and grounding. Some two-stage approaches use different training objectives between the first and second stages. Li et al. (2022); McKinzie et al. (2024); Deshmukh et al. (2023); Li et al. (2024b) use contrastive loss in the first stage to align different modalities. During this process, modality encoders are trained to produce aligned embeddings. In the second stage, the language modeling loss is used to integrate these aligned representations with the LLM.

Reconstruction loss is also used for modality integration (Ge et al., 2023; Yang et al., 2024; Pan et al., 2024; Zhu et al., 2024c). These studies aim to reconstruct non-text modalities as the output of modality integration. In the cross-modality reconstruction process using LLMs, the LLM creates conditions for reconstruction. The goal is to generate appropriate conditions so that outputs can be reconstructed by understanding the context contained in cross-modality data. Additionally, in the vision-language field, DICE loss is sometimes used for segmentation performance (Lai et al., 2024; Gu et al., 2024; Zhang et al., 2024f).

6 Challenges and Future direction

In our survey of MLLMs, we introduce several key research challenges and promising future research directions. MLLMs face several technical limitations including hallucination problems (Xu et al., 2024a), inconsistencies between modalities, and vulnerability to adversarial attacks (Shayegani et al., 2023; Zhao et al., 2023; Jeong, 2024). Recent work has highlighted these issues, showing how carefully crafted inputs can exploit weaknesses in multimodal processing.

Based on our analysis, we identify several promising research directions. First, incorporating Retrieval-Augmented Generation(RAG) into multimodal systems shows potential for improving factual accuracy. Second, developing multimodal agents with enhanced reasoning capabilities represents an important frontier. Third, current MLLMs lack persistent memory mechanisms beyond context windows—an essential capability for more general intelligence that would allow models to leverage previous interactions without explicit context injection. Finally, while reasoning-focused LLMs (OpenAI et al., 2024b; DeepSeek-AI et al., 2025; Yang et al., 2025) have made significant progress, their multimodal counterparts remain underdeveloped in open-source research, presenting a critical opportunity. Addressing these challenges will advance the development of more robust and capable multimodal systems.

7 Conclusion

Multimodal LLMs extend language models by integrating visual, audio, and other modalities through abstraction, projection, semantic embedding and cross-attention on different fusion levels. We introduce a clear taxonomy based on three dimensions: Architectural strategies for Modality fusion, representation paradigms(joint, coordinated, hybrid), and training paradigms(single-stage, two-stage, multi-stage). By reviewing 125 MLLMs, we reveal the intentions and design philosophies of researchers, showing how they approached challenges in effectively and efficiency. We examine ongoing challenges in evaluation, multimodal hallucination, and persistent memory, while highlighting promising research directions such as retrieval-augmented generation and reasoning-focused MLLMs. This survey provides future researchers with insights into the evolution of design choices and research directions, offering a roadmap that builds upon previous work to guide the development of more robust and efficient multimodal language models.

8 Limiation

This survey analyzes MLLM architectures and methods qualitatively, but has several limitations. First, cross-modal evaluation remains difficult due to the lack of standardized metrics across different modalities. The absence of common benchmarks hinders effective comparison between models, obscuring performance differences. Therefore, we do not provide new experiments or direct comparisons between models. We rely on findings reported in original papers. Second, many papers implementing specific integration mechanisms do not clearly state their design motivations—this is particularly problematic since the same architectural components often serve different functional purposes across studies. Thus, we focused on models with clear descriptions of their integration mechanisms(projection layers, cross-attention, etc.) and design choices. Models with unclear architectural details are excluded from our analysis tables. Third, we examine MLLMs with direct integration between modalities and the LLM backbone. We do not cover indirect integration approaches, such as LLMs connecting to other modalities through intermediate models(like diffusion models) with shared loss functions. Lastly, most surveyed MLLMs use language modeling loss for training. The lack of models using different loss functions limits our ability to compare diverse training objectives. This reflects current research trends rather than an intentional exclusion.

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A related survey

This appendix compares nine representative MLLM surveys across four key dimensions: (1) Semantic Fusion Mechanisms (Projection, Abstraction, Cross-attention, Semantic Embedding), (2) LLM-relative Fusion Levels (Early, Intermediate, Hybrid), (3) Data & Training Paradigms (Contrastive, Generative, Hybrid), and (4) Representation Learning Approaches (Joint vs. Coordinated). None of the existing surveys covers all four dimensions cohesively. For example, Shukang Yin et al. (2023) introduces token vs. feature fusion but lacks semantic mechanism analysis and representation taxonomy. Li and Tang (2024) proposes data- and feature-level fusion but does not classify LLM-centric fusion levels or representation approaches. Caffagni et al. (2024) reviews two-stage training but omits detailed loss function comparisons and representation learning frameworks. Jiang et al. (2025) offers a unified modeling perspective but does not systematically examine fusion levels or training paradigms. Surveys by Wu et al. (2023), Jin et al. (2024), Zhang et al. (2024b), and Song et al. (2025) each cover select aspects but do not provide the integrated view across all four dimensions. Our work addresses this gap by offering a unified framework along these dimensions (see Table 2).

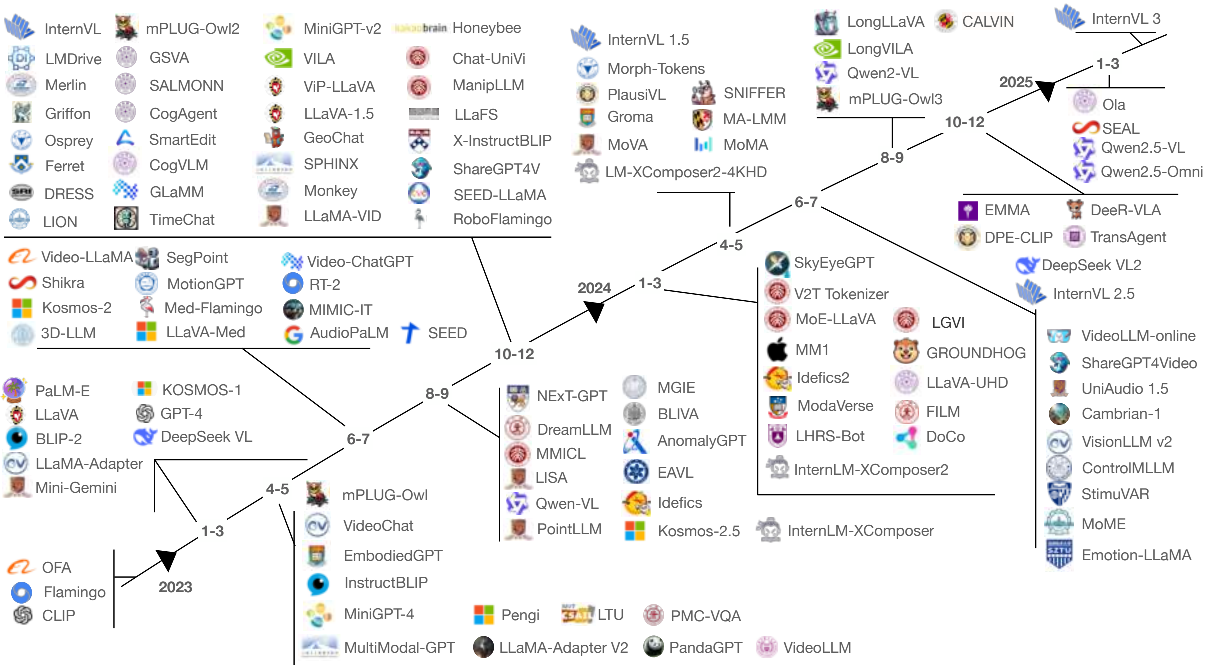

Figure 3: Timeline of major developments in multimodal architectures

Table 2: Comparison with existing MLLM surveys based on our proposed classification dimensions. Coverage is indicated by symbols (✓: Covered, ×: Not Covered, ∼: Partial/Different). Columns represent cited surveys (ordered based on new data, mostly chronological) and the current work (Ours).

Dimension / Survey	Shukang Yin et al. (2023)	Wu et al. (2023)	Caffagni et al. (2024)	Jin et al. (2024)	Li and Tang (2024)	Zhang et al. (2024b)	Han et al. (2025)	Jiang et al. (2025)	Song et al. (2025)	**Ours**
(1) Semantic Int. Mech.	∼(Fig 2)	×	∼(Sec 2.3)	∼(Sec 2.2)	×	∼(Sec 2.2)	×	×	∼(Sec 3,4)	✓
(2) Fusion Level (LLM-relative)	∼(Diff.fusion levels)	×	×	×	∼(Diff. fusion levels)	×	∼(no classification applied)	×	×	✓
(3) Data & Training Paradigm	×	×	×	×	×	×	×	×	×	✓
(4) Rep. Learning Approach	×	×	×	×	×	×	×	×	×	✓

Model	Abstractor Layer	Projection Layer	Semantic Embedding Layer	Cross-attention Layer	Fusion level	Representation
EmbodiedGPT(Mu et al., 2023)	-	Linear	Q-former	-	Early	Joint
Flamingo(Alayrac et al., 2022)	Perceiver Resampler	-	-	Within Model	Inter	Joint
Kosmos-1(Huang et al., 2023)	Perceiver Resampler	MLP(last layer of ViT)	-	-	Early	Joint
Idefics2(Laurençon et al., 2024)	Perceiver Resampler	MLP	-	-	Early	Joint
RoboFlamingo(Li et al., 2023c)	Perceiver Resampler	-	Cross-attention(within Model)	-	Inter	Joint
LLaVA-UHD(Xu et al., 2024c)	Perceiver Resampler	-	-	-	Early	Joint
DeeR-VLA(Yue et al., 2024)	Perceiver Resampler	-	-	-	Inter	Joint
BLIP-2(Li et al., 2023b)	-	Linear	Q-former	-	Early	Hybrid
MM1(McKinzie et al., 2024)	C-Abstractor	-	-	-	Early	Joint
LLaVA-1.5(Liu et al., 2024)	-	MLP	-	-	Early	Joint
InstructBLIP(Dai et al., 2023)	-	Linear	Q-former	-	Early	Joint
CogVLM(Wang et al., 2024b)	-	MLP	-	within Model	Inter	Joint
CogAgent(Hong et al., 2024)	-	MLP	-	within Model	Hybrid	Joint
mPLUG-Owl2(Ye et al., 2024c)	VIsual Abstractor	-	-	Modality Adaptive Module	Inter	Joint
mPLUG-Owl3(Ye et al., 2024a)	-	Linear	-	within Model(Hyper Attention Transformer block)	Inter	Joint
LLaMA-VID(Li et al., 2024c)	-	Linear	-	external layer	Early	Joint
Video-LLaMA(Zhang et al., 2023b)	Q-former	Linear	-	-	Early	Joint
SALMONN(Tang et al., 2024)	Q-former	-	-	-	Early	Joint
LMDrive(Shao et al., 2024)	Q-former	MLP	-	-	Early	Joint
CALVIN(Somepalli et al., 2024)	Q-former	Linear	-	-	Early	Joint
Merlin(Yu et al., 2024)	Convolution	-	-	-	Early	Joint
SEAL(Sun et al., 2025)	Convolution	MLP	-	-	Early	Coord
LLaMA-Adapter V2(Gao et al., 2023)	-	Linear	-	within Model(self attention layer)	Inter	Joint
VideoChat(K.-Y. R. Li et al., 2023)	Q-former	Linear	-	-	Early	Joint
LLaVA-Med(Li et al., 2023a)	-	Linear	-	-	Early	Joint
VILA(Lin et al., 2024b)	-	Linear / Transformer	-	-	Early	Joint
PandaGPT(Su et al., 2023)	-	Linear	-	-	Early	Joint
SEED(Ge et al., 2023)	-	Linear	Q-former	-	Early	Hybrid
MoE-LLaVA(Lin et al., 2024a)	-	Linear	Q-former	-	Early	Joint
VisionLLM(Wang et al., 2023)	-	Transformer	-	-	Early	Joint
ManipLLM(Li et al., 2024a)	-	Transformer	-	within Model(self attention layer)	Hybrid	Joint
LongVILA(Chen et al., 2024b)	-	Linear / Transformer	-	-	Early	Joint
PMC-VQA(Zhang et al., 2024d)	-	MLP / Transformer	-	-	Early	Joint
X-InstructBLIP(Panagopoulou et al., 2024)	-	Q-former + Linear	-	-	Early	Joint
VisionLLM v2(Wu et al., 2024)	-	Q-former + Linear	-	-	Early	Joint
MiniGPT-4(Zhu et al., 2024a)	-	Linear	-	-	Early	Joint
TimeChat(Ren et al., 2024)	-	Linear	Q-former	-	Early	Joint
MA-LMM(He et al., 2024)	-	-	Q-former	-	Early	Joint
LION(Chen et al., 2024a)	-	MLP	Q-former	-	Early	Joint
Sniffer(Qi et al., 2024)	-	-	Q-former	-	Early	Joint
PlausiVL(Mittal et al., 2024)	-	Linear	Q-former	-	Early	Hybrid
VideoLLM-online(Chen et al., 2024a)	-	MLP	-	-	Early	Joint
Beyond Text(Zhu et al., 2024c)	-	Linear	-	-	Early	Joint
RSGPT(Hu et al., 2023)	-	-	Q-former	-	Early	Joint
LHRS-Bot(Muhtar et al., 2024)	-	-	Perceiver Resampler	-	Early	Joint
AnomalyGPT(Gu et al., 2024)	-	Linear	Convolution	-	Early	Hybrid
Cambrian-1(Tong et al., 2024)	-	-	Cross-attention(like Convolution)	-	Inter	Joint
MultiModal-GPT(Gong et al., 2023)	Perceiver Resampler	-	-	Within Model	Inter	Joint
Med-Flamingo(Moor et al., 2023)	Perceiver Resampler	-	-	Within Model	Inter	Joint
RoboFlamingo(Li et al., 2023c)	Perceiver Resampler	-	-	Within Model	Inter	Joint

Table 3: Comparison of 50 Multi-Modal Architectures.